AUTOMATIC WATER LEVEL INDICATOR AND CONTROLLER USING ARDUINO

A Real-Time / Field-Based research project report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

Submitted by

KADARI VENU (23B81A0460)

K.SRINIVASULU (23B81A0450)

V.RAHUL KUMAR REDDY (23B81A0435)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

CVR COLLEGE OF ENGINEERING

(An Autonomous Institution with Grade 'A' & Affiliated to JNTUH)
Mangalpalli (V), Ibrahimpatnam (M), Ranga Reddy (D), Telangana – 501510

2024-25

AUTOMATIC WATER LEVEL INDICATOR AND CONTROLLER USING ARDUINO

A Real-Time / Field-Based research project report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

Submitted by

KADARI VENU (23B81A0460)

K.SRINIVASULU (23B81A0450)

V.RAHUL KUMAR REDDY (23B81A0435)

Under the Supervision of

Ms. S.SOUJANYA

Assistant Professor of ECE

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

CVR COLLEGE OF ENGINEERING

(An Autonomous Institution with Grade 'A' & Affiliated to JNTUH)
Mangalpalli (V), Ibrahimpatnam (M), Ranga Reddy (D), Telangana – 501510

2024-25

Cherabuddi Education Society's
CVR COLLEGE OF ENGINEERING
(An Autonomous Institution)
ACCREDITED BY NATIONAL BOARD OF ACCREDITATION, AICTE
(Approved by AICTE & Govt. of Telangana and Affiliated to JNT University)
Vastunagar, Mangalpalli (V), Ibrahimpatan (M), R.R. District, PIN - 501 510
Web : http://cvr.ac.in, email : info@cvr.ac.in
Ph : 08414 - 252222, 252369, Office Telefax : 252396, Principal : 252396 (O)

-

Certificate

This is to certify that the Real-Time / Field-Based research project titled **"AUTOMATIC WATER LEVEL INDICATOR AND CONTROLLER USING ARDUINO"** submitted to **CVR College of Engineering,** affiliated to **JNTUH,** by **KADARI VENU (23B81AA0460),K.SRINIVASULU (23B81A0450), V.RAHULKUMARREDDY (23B81A0435),** is a bonafide record of the work done by the students towards partial fulfillment of requirements for the award of the degree of **Bachelor of Technology in Electronics & Communication Engineering** during the academic year 2024-2025.

*

Supervisor

Ms.S.Soujanya
Assistant professor of ECE

Head of the Department

Dr. P. Srinivasa Rao
Professor and Head of ECE

Date:

Place:

Declaration

I hereby declare that the Real-Time / Field-Based research project report titled "**AUTOMATIC WATER LEVEL INDICATOR AND CONTROLLER USING ARDUINO**" submitted to Department of Electronics and Communication Engineering, CVR College of Engineering is a record of original work done by us under the guidance of **Ms.S.SOUJANYA**. The information and data given in the report is authentic to the best of my knowledge. This project report is not submitted to any other university or institution for the award of any degree or diploma or published any time before.

KADARI VENU(23B81A0460)

K.SRINIVASULU (23B81A0450)

V.RAHUL KUMAR REDDY (23B81A0435)

Date:

Place:

Acknowledgements

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose encouragement and guidance has been a source of inspiration throughout the course of the project.

We would like to express our sincere gratitude to our supervisor**, Ms.S.Soujanya, Assistant professor**, **ECE Department**, **CVR College of Engineering,** whose guidance and valuable suggestions have been indispensable to bring about the successful completion of our project.

We wish to acknowledge special thanks to the Project Coordinators **Mr. Karrar Hussain,** Associate Professor of ECE**,** and **Ms. J. Jyostna,** Sr. Assistant Professor of ECE, for assessing seminars and giving us valuable suggestions for our project.

We would like to express our sincere thanks to our Head of the ECE Department, **Dr. P. Srinivasa Rao,** for his continuous moral support in completing our project.

It is great pleasure to convey our profound sense of gratitude to our principal **Dr. K**. **Ramamohan Reddy**, CVR college of Engineering for having been kind enough for arranging the necessary facilities for executing the project in the college.

We would also like to express our gratitude to all the staff members and lab faculty, Department of Electronics and Communication Engineering, CVR College of Engineering for their constant help and support.

We wish a deep sense of gratitude and heartfelt thanks to management for providing excellent lab facilities and tools. Finally, we thank all those whose guidance helped us in this regard.

Abstract

This project presents the design and implementation of a water level indicator and controller system using an Arduino microcontroller. The primary goal is to monitor and control the water level in a storage tank to prevent both overflow and dry-run conditions. The system utilizes water level sensors to detect different levels within the tank and displays real-time status on an LCD. When the water drops below a predefined minimum level, the Arduino automatically activates a pump, and when the tank reaches the maximum level, it switches the pump off. This ensures efficient water usage, minimizes wastage, and protects the pump from running dry, ultimately conserving energy and resources.

The Arduino-based water level controller offers a low-cost, reliable, and user-friendly solution for residential, agricultural, and industrial water management. The integration of basic electronic components such as sensors, relays, buzzers, and an LCD display enables the system to function independently with minimal human intervention. Additionally, the modular nature of the design allows for future upgrades, such as wireless monitoring or smartphone integration. This project demonstrates the effective application of embedded systems in addressing real-world challenges in resource conservation and automation.

Keywords:

- **Automatic Pump Control**
- **Sensor-Based System**
- **Water Tank Monitoring**
- **Energy Conservation**
- **Home Automation**
- **Smart Water Management**

IoT Integration (optional, for future enhancement)

Contents

Acknowledgements i

Abstract ii

Contents iii

List of Figures vi

List of Tables vii

Chapter 1 Introduction

1.1 Introduction 10

1.2 Aim of the Project 10

1.3 Literature survey 10

1.4 Technical approach 11

1.5 Application of the project 11

1.6 Organization of the project 12

1.7 Hardware Description 12

1.8 project Description 13

Chapter 2 Fundamental of Embedded systems

2.1 Embedded system basics 14

2.2 Embedded system software 15

2.3 Embedded system design tools 16

2.4 Microcontrollers 17

2.4.1 Types of Microcontrollers 17-19

2.4.2 How Microcontrollers Works 20

2.4.3 Uses of Microcontrollers 21

2.4.4 Application of Microcontrollers 22

Chapter 3 Hardware Description

3.1 List of materials used 23

3.2 Arduino UNO 23-25

3.3 Ultrasonic sensor 26-28

3.4 Lcd display 29-30

3.5 Relay module 31-32

3.6 pump 33

Chapter 4 Project Description

4.1 Objectives 34

4.2 Circuit diagram 34

4.3 Code 35-37

4.4 Flow Chart 38

4.5 Applications 38-39

Chapter 5 Result and Discussion 40-41

Chapter 6 Conclusion and Future scope 42

List of Figures

Fig.2.1. ARM Microcontroller 18

Fig.2.2. AVR Micro Controller 18

Fig.2.3 FPGA-based Microcontroller 19

Fig.2.4 Working of Microcontroller 21

Fig.3.1 Arduino UNO 23

Fig.3.2 Arduino UNO pins 24

Fig.3.3 Ultrasonic Sensor 26

Fig.3.4 Distance Calculation 27

Fig.3.5. Detection of Liquid level 29

Fig.3.6 LCD Display 30

Fig.3.7 I2C Module 31

Fig.3.8 Relay Module 31

Fig.3.9 Pump 33

Fig.4.1 circuit diagram 34

Fig.4.2 Flow chart 38

Fig.5.1 water level indicator and controller 40

List of Tables

Table.3.1. Modulation specification 28

Table.3.2. Hardware information 28

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

This project presents a **Water Level Indicator and Controller** system using **Arduino**, which automates the process of monitoring and controlling the water level in a tank. The system employs an **ultrasonic sensor** to measure the water level, displaying real-time readings on an **LCD screen**. When the water drops below a predefined level, the **Arduino** activates a **water pump** through a **relay module**, filling the tank until the desired level is reached.

1.2 AIM OF PROJECT

The aim of this project is to design and implement an automatic water level monitoring and control system using Arduino UNO

1.3 LITERATURE SURVEY

Water level monitoring and control systems have been an area of interest for researchers and engineers due to the increasing demand for efficient water management in domestic, agricultural, and industrial sectors. Various techniques and technologies have been explored over the years to automate this process. This literature survey highlights key developments and existing systems related to water level indicators and controllers, particularly those utilizing microcontrollers like Arduino.

LCD and Relay Integration

Display units like **16x2 LCDs** are commonly used for providing user-friendly interfaces in water level systems. Similarly, **relay modules** allow safe operation of high-power devices like water pumps, enabling seamless automation. Research suggests that using relays with Arduino ensures both safety and efficienc1y in pump control.

Energy and Resource Efficiency

Studies have shown that automated water level controllers significantly reduce electricity and water wastage by ensuring the pump operates only when necessary. This results in lower utility bills and reduced environmental impact.

1.4 TECHNICAL APPROACH

The technical approach of this project involves designing and implementing a fully automated system that can monitor and control the water level in a tank using an Arduino microcontroller and various interfaced components. The system operates based on real-time data from an ultrasonic sensor, processes it through the Arduino, and performs corresponding actions such as displaying the level and controlling a water pump.

1.5 APPLICATION OF THE PROJECT

The Water Level Indicator and Controller system has a wide range of practical applications across domestic, commercial, and industrial sectors. Some key applications include:

1. Residential Water Tanks

- Automatically monitors and controls water levels in overhead or underground tanks.
- Prevents overflow and dry-run of water pumps.
- Saves electricity and water by ensuring efficient pump usage.

2. Agricultural Irrigation Systems

- Maintains optimal water levels in irrigation tanks or reservoirs.
- Ensures timely water supply to crops without manual supervision.

3. Industrial Water Storage

- Useful in industries where continuous water supply is essential for operations or cooling systems.
- Automates water refilling and reduces manual effort.

4. Commercial Buildings and Apartments

- Manages water distribution across multiple tanks or floors.
- Reduces maintenance costs and prevents water wastage.

5. Water Treatment Plants

- Controls levels in processing and storage tanks.
- Supports automation in filtration and chemical dosing units.

6. Rainwater Harvesting Systems

- Monitors harvested water storage levels.
- Can control diversion to overflow or secondary storage tanks.

1.6 ORGANIZATION OF THE PROJECT

FUNDAMENTALS OF EMBEDDED SYSTEM

The fundamentals of embedded systems encompass the design and implementation of specialized computing systems that perform dedicated functions within larger mechanical or electrical systems. These systems are characterized by their integration of hardware and software, tailored to specific applications such as automotive control, industrial automation, medical devices, and consumer electronics. Key components include microcontrollers or microprocessors, memory, input/output interfaces, and real-time operating systems (RTOS) to manage tasks and ensure timely responses. Embedded systems prioritize reliability, efficiency, and minimal resource consumption, often operating under stringent constraints in terms of power, size, and cost. Understanding these fundamentals is crucial for developing robust, efficient, and responsive embedded solutions tailored to specific operational requirements.

1.7 HARDWARE DESCRIPTION

Hardware Specifications:

- Arduino UNO
- Ultrasonic Sensor
- Relay
- 16x2 LCD

Software Specifications:

- Arduino IDE

1.8 PROJECT DESCRIPTION

- The **Water Level Indicator and Controller using Arduino** is a microcontroller-based automation system designed to **monitor and control the water level in a storage tank**. The system aims to minimize water wastage, reduce manual monitoring, and prevent issues such as tank overflow and dry running of the water pump.
- This system uses an **ultrasonic sensor (HC-SR04)** to detect the water level by measuring the distance between the sensor (mounted on the tank lid) and the water surface. The data from the sensor is continuously sent to an **Arduino Uno**, which processes the input and determines the current water level.
- The system is programmed to trigger a **relay module** to automatically turn ON the **water pump** when the water level falls below a certain minimum threshold. Similarly, when the water level reaches a maximum threshold (tank full), the pump is turned OFF to prevent overflow.
- To make the system user-friendly, a **16x2 LCD display** is connected to show the current water level and the status of the pump (ON/OFF). This provides real-time visual feedback for users.

BENEFITS:

- **Saves water by preventing overflow.**
- **Reduces electricity usage and increases pump lifespan.**
- **Eliminates the need for manual water level checking.**
- **Low-cost and easy-to-implement solution for homes, apartments, farms, and industries.**

CHAPTER 2

FUNDAMENTALS OF EMBEEDED SYSTEMS

2.1 Embedded systems basics

It may be asked what is an embedded system. With many processor based systems and computers it is useful to define what an embedded system is. A convenient definition for an embedded system is: An embedded system is any computer system contained within a product that is not described as a computer.

Using this embedded system definition it is possible to understand the various basic characteristics one. Typically they are:

- Embedded systems are designed for a specific task. Although they use computer techniques, they cannot be used as a general purpose computer using a variety of different programmes for different task. In this way their function can be focussed onto what they need to do, and they can accordingly be made cheaper and more efficiently.
- The software for embedded systems is normally referred to as firmware. Rather than being stored on a disc, where many programmes can be stored, the single programme for an embedded system is normally stored on chip and it is referred to as firmware Embedded systems contain two main elements
- Embedded system hardware: As with any electronic system, an embedded system requires a hardware platform on which to rum. The hardware will be based around a microprocessor or microcontroller. The embedded system hardware will also contain other elements including memory, input output (1/0)
- interfaces as well as the user interface, and the display
- Embedded system software : The embedded system software is written to perform a particular function. It is typically written in a high level Format and then compiled down to provide code that can be lodged within a non-volatile memory within the hardware.

Embedded systems hardware

When using an embedded system there is a choice between the use of a microcontroller or microprocessor

- Microcontroller based systems: A microcontroller is essentially a CPU, central processor unit, or processor with integrated memory or peripheral devices. As fewer external components are needed, embedded system ng microcontrollers tend to be more widely used
- Microprocessor based systems: Microprocessor contain a CPU but me external chips for memory and peripheral interfaces. As they require more devices on the board, but they allow mere expansion and selection of exact peripherals, this approach tends to be used for the larger embedded systems

Whatever type of processor is used in the embedded system, it may be a very general purpose type of one of the many highly processors intended for a particular application. In some cases custom designed chips may be viable for a particular application if quantities are sufficiently high. One common example of a standard class of dedicated processor is the digital signal processor, DSP. This type of processer is used for processing audio and image files in particular. Processing is required very quickly as they may be used in applications such as mobile phones and the like.

2.2 Embedded systems software

One of the key elements of any embedded system is the software that in used to non the microcontroller

There is a variety of ways that this can be written:

Machine code: Machine code is the most basic code that is used for the processor unit. The code is normally in hex code and provides the basic instructions for each operation of the processor This form of code is rarely used for embedded systems these days

Programming language: Writing machine code is very laborious and time consuming. It is difficult to understand and debug. To overcome this, high level programming languages are often used. Languages including C, C++, de are commonly used.

The code for the embedded system will typically be stored on a form of mm-volatile memory held on the processor brand. The code is called firmware-the idea is that it is not updated in the same way that software is, being held in the embedded system and it cannot be changed by the user. Often it is possible to update the software, but this can mean changing the memory card on which the firmware is held, er by updating it in another way,

Often additional tools may be used to help with the development of the firmware. Offen programming can become complicated and it is necessary to ensure the firm warm for the embedded system operates correctly.

2.3 Embedded systems design tools

Many embedded systems are complicated and require large levels of software for them to operate.

Developing this software can be timing consuming, and it has to be very accurate for the embedded system to operate correctly. Coding in embedded systems is one of the main areas where faults occur

To help simplify the process, software development tools are normally used. These help the software developer to program more quickly, and also more accurately.

When developing an embedded system, one of the options is to base the computational hardware around a microcontroller, MCU rather than a microprocessor, MPU.

Both approaches have their attractions, but generally they will be found in different applications. Typically the microcontroller, MCU, is found in application where size, low power and low cost are key requirements.

The MCU, microcontroller is different to a micro processer in that it contains more elements of the overall processing engine within the one chip.

Bringing most of the processing engine components onto a single chip reduces size and cost This enables it to become economical viable to digitally control even more devices and processes.

Alan it is found that mixed signal microcontrollers are being increasingly used, integrating analogue components needed to control non-digital electronic systems.

2.4 Microcontrollers

A microcontroller is a small, low-cost computer-on-a-chip that is designed to perform a specific set of tasks. It is used in a variety of applications, including controlling machines, sensing and monitoring devices, and automated systems. A microcontroller typically consists of a central processing unit (CPU), memory, Input/output (100) ports, and support for various peripherals such as tinters, counters, and analog-to-digital converters. It is programmed using a specialized computer language and is capable of executing instructions stored in its memory Some common examples of microcontrollers include the Arduino, Raspberry Pi and the PIC microcontroller-

2.4.1 Types of Microcontroller:

Here are some of the most common types of microcontrollers:

8-bit Microcontrollers: These are the most basic type of microcontrollers, typically used in simple applications such as toys, small appliances, and remote controls. They have a limited processing power and memory capacity, but they are easy to use and cost-effective

16-bit Microcontrollers: These are more advanced than Il-bit microcontrollers and are capable of performing more complex tasks. They are commonly used in applications such as medical devices, automotive systems, and industrial control systems.

32-bit Microcontrollers: These are the most powerful and feature-rich micro controller, capable of handling large amounts of data and performing highspeed processing. They are used in applications such as gaming systems, multimedia devices, and high-end industrial automation.

ARM Microcontrollers: These microcontrollers are based on the ARM architecture and are widely used in a variety of applications, including mobile devices, automotive, and industrial control system

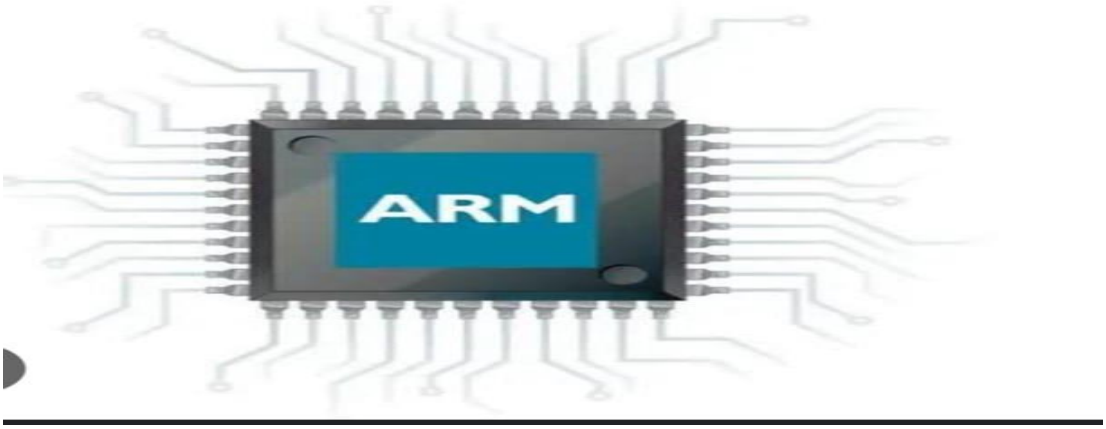

Fig.2.1.ARM Microcontroller

PIC Microcontrollers: These microcontrollers are mumsfactured by Microchip Technology and are commonly used in a wide range of applications, including home appliances, automotive systems, and medical devices.

AVR Microcontrollers: These microcontrollers are manufactured by Atinel Corporation and are commonly used in applications such as robotics, industrial control systems, and consumer electronics.

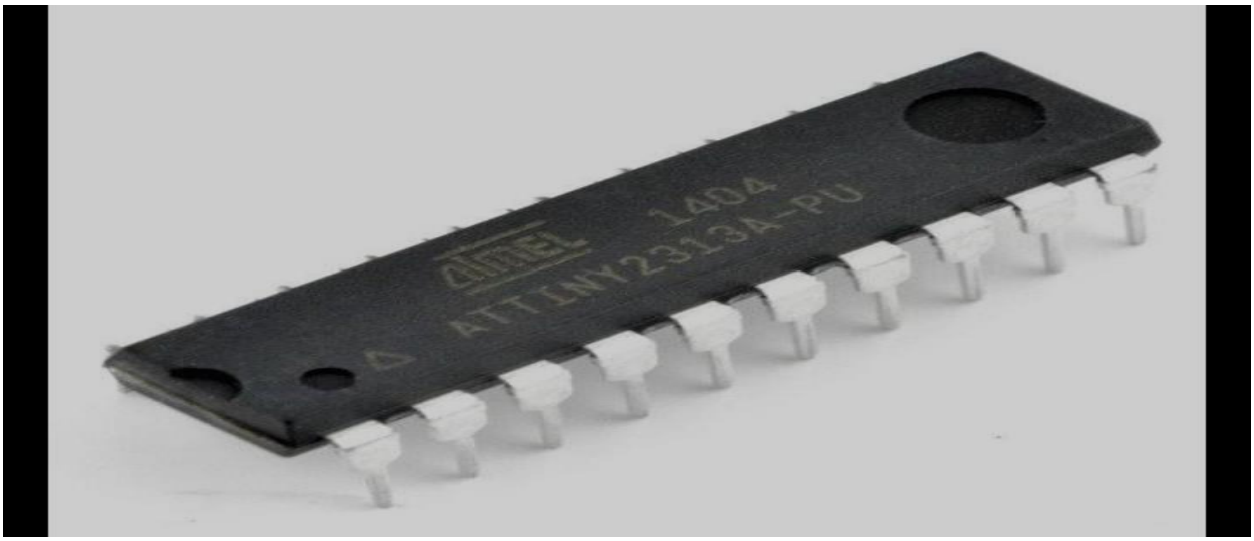

Fig.2.2.AVR Microcontroller

CPU: The microcontroller is referred to has CPU device since it is utilized to carry and decode a data before effectively completing the signal duty. All microcontroller components are connected to a specific system utilizing a central processing unit. The CPU can decode instructions retrieved from the programmable memory

Memory: The memory chip of a microcontroller similarly to a microprocessor in that it stores all data as well as programming. Microcontrollers have a limited quantity of RAM/ROM ,flash memory for storing program source code

Fig.2.3.FPGA-based Microcontroller

FPGA-based Microcontrollers: An **FPGA-based microcontroller** is a system where the **core processing logic** of a microcontroller is **implemented (or emulated) inside an FPGA chip**. Instead of having a fixed architecture like traditional MCUs (e.g., Arduino, STM32, PIC), the logic of the MCU is defined using **hardware description languages (HDLs)** like VHDL or Verilog, and then programmed onto the FPGA.

Input and Output ports: in general, these ports are made to interface or otherwise drive various appliances like LED, LCD, printer, and so on.

ADC (Analog to Digital Converter) : An **Analog-to-Digital Converter (ADC)** is a built-in module in many microcontrollers that **converts analog signals (continuous voltage levels) into digital values (binary numbers)**. This allows the microcontroller to read and process real-world analog inputs like **temperature, light, sound, or any voltage-based sensor data**.

Control Interpretation: **Control interpretation** in a microcontroller refers to how the microcontroller **understands and responds to control instructions**—whether those come from code, sensors, user input, or communication interfaces. It's essentially the process of **interpreting commands and executing control logic** to drive outputs or take decisions in real-time.

2.4.2 How Microcontrollers Work?

A microcontroller is a compact integrated circuit designed to govern specific operations in an embedded system. It works like a tiny computer on a single chip, containing a processor (CPU), memory (RAM and Flash/ROM), and input/output (I/O) peripherals. The microcontroller operates by executing a program stored in its memory. When powered on, it starts reading inputs from connected devices like sensors or buttons, processes the data according to the programmed instructions, and then sends appropriate outputs to devices such as LEDs, motors, or displays. For example, in a temperature monitoring system, the microcontroller can read temperature data from a sensor, compare it with a predefined threshold, and turn on a fan if the temperature is too high. Microcontrollers are used in everyday devices like washing machines, microwave ovens, remote controls, and even cars, making them essential in the world of automation and control systems.

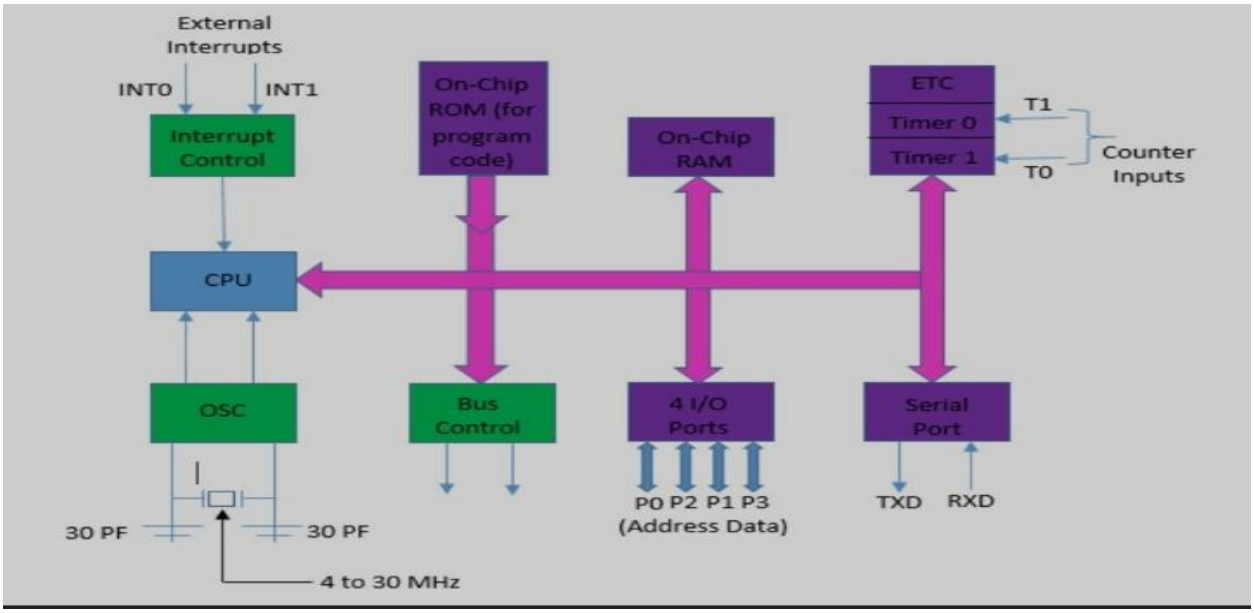

Fig.2.4.Working of Microcontrollers

2.4.3 USES OF MICROCONTROLLER:

Microcontrollers are compact integrated circuits designed to govern a specific operation in an embedded system. They are widely used across various fields due to their small size, low cost, and efficient control capabilities.

Here are **some key uses of microcontrollers**:

Consumer Electronics

- **Washing machines**, **microwaves**, **refrigerators** – to manage operations like timers, sensors, and motor control.
- **Remote controls**, **TVs**, **audio systems** – for interface and communication control.
- **Digital cameras** – for auto-focus, exposure settings, and image processing.

Automotive Applications

- **Engine control systems** – fuel injection, ignition timing, emission control.
- **Airbags and ABS** – real-time monitoring and triggering.

- **Dashboard instruments** – speedometer, fuel gauges, digital displays.

Industrial Automation

- **Robotics** – controlling sensors, motors, and decision logic.
- **Process control systems** – temperature, pressure, and flow monitoring.
- **CNC machines** – for precision movement and automation.

Medical Devices

- **Pacemakers**, **glucometers**, **digital thermometers** – for data collection and processing.
- **Infusion pumps** – for controlled delivery of medication.

2.4.4 APPLICATIONS OF MICROCONTROLLERS

- **Digital Thermometers – Measures and displays body temperature. Washing Machines – Controls wash cycles, water levels, and timers.**
- **Microwave Ovens – Manages cooking time, power levels, and user interface.**
- **Air Conditioners – Regulates temperature, fan speed, and sleep modes.**
- **Digital Cameras – Handles focus, exposure, and image processing.**
- **Smart TVs – Manages input/output interfaces and user settings.**
- **Remote Controls – Encodes and transmits signals to control devices.**
- **Automatic Street Lights – Turns lights on/off based on time or light sensors.**
- **Electronic Voting Machines (EVMs) – Ensures secure vote casting and counting.**
- **Car Engine Control Unit (ECU) – Manages ignition, fuel injection, and emissions.**
- **Anti-lock Braking System (ABS) – Monitors wheel speed and controls braking.**
- **Pacemakers – Maintains and regulates heartbeat.**
- **Traffic Light Controllers – Controls timing and sequences of lights.**
- **Industrial Robots – Performs tasks like welding, painting, and assembly.**
- **Home Automation Systems – Controls lights, fans, and security devices remotely.**
- **Smart Energy Meters – Measures and reports electricity usage.**
- **Drones – Manages flight control, GPS navigation, and stabilization.**
- **3D Printers – Controls motors, temperature, and print precision.**
- **Security Systems – Manages cameras, alarms, and motion sensors.**
- **Automatic Plant Watering Systems – Detects soil moisture and activates watering**

CHAPTER 3

HARDWARE DESCRIPTION

3.1 LIST OF MATERIALS USED

1)ARDUINO UNO

2) ULTRASONIC SENSOR

3)LCD DISPLAY

4)RELAY

5)PUMP

3.2 ARDUINO UNO

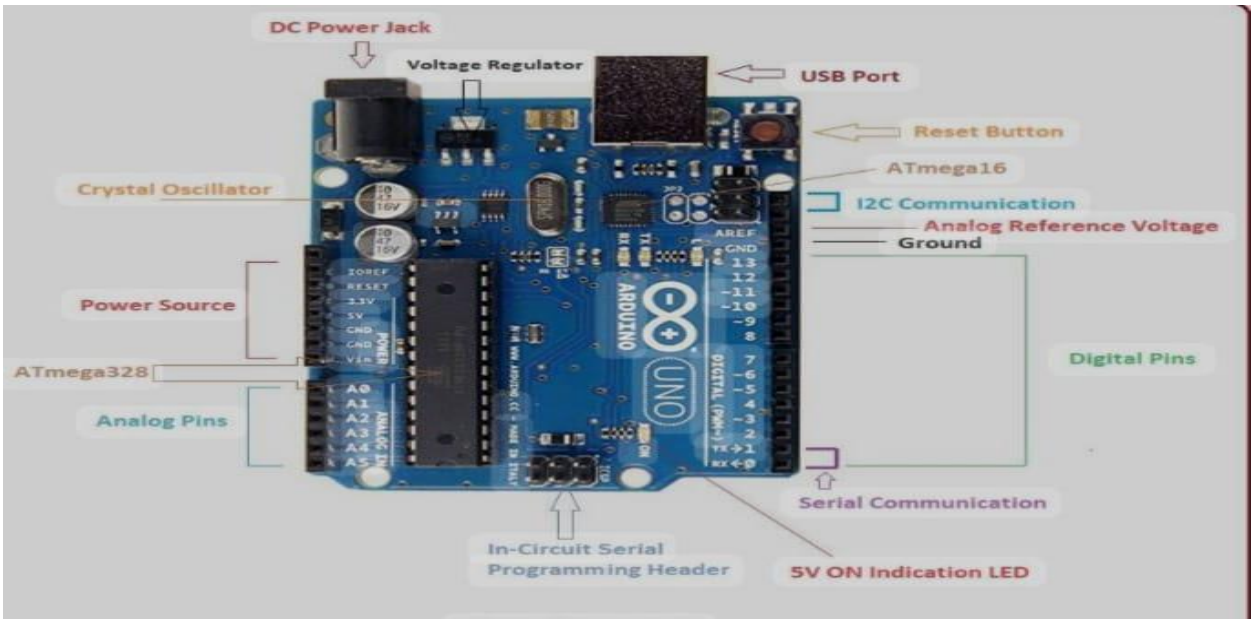

Fig.3.1.Arduino UNO

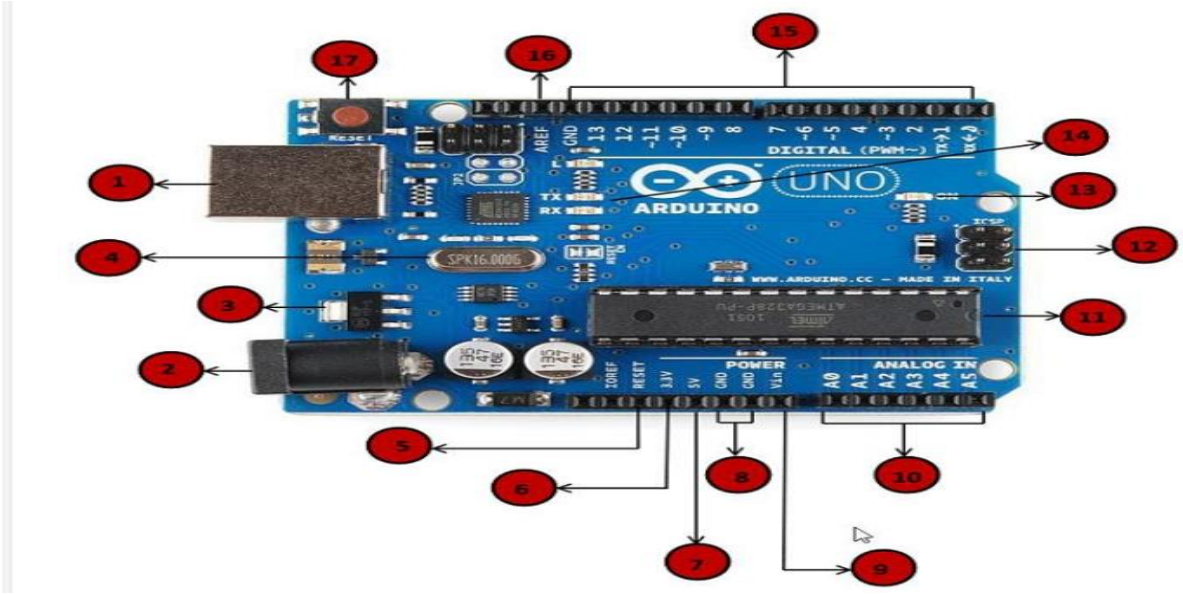

Fig.3.2.Arduino Uno pins

1.Power USB

Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection (1).

2.Power (Barrel Jack)

Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack (2).

3.Voltage Regulator

The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.

4.Crystal Oscillator

The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16,000,000 Hertz or 16 MHz.

5,17.Arduino Reset

You can reset your Arduino board, i.e., start your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button (17) on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET (5).

6,7,8,9.Pins (3.3, 5, GND, Vin)

- 3.3V (6) − Supply 3.3 output volt
- 5V (7) − Supply 5 output volt
- Most of the components used with Arduino board works fine with 3.3 volt and 5 volt.
- GND (8)(Ground) − There are several GND pins on the Arduino, any of which can be used to ground your circuit.
- Vin (9) − This pin also can be used to power the Arduino board from an external power source, like AC mains power supply.

10.Analog pins

The Arduino UNO board has six analog input pins A0 through A5. These pins can read the signal from an analog sensor like the humidity sensor or temperature sensor and convert it into a digital value that can be read by the microprocessor.

11.Main microcontroller

Each Arduino board has its own microcontroller (11). You can assume it as the brain of your board. The main IC (integrated circuit) on the Arduino is slightly different from board to board. The microcontrollers are usually of the ATMEL Company. You must know what IC your board has before loading up a new program from the Arduino IDE. This information is available on the top of the IC. For more details about the IC construction and functions, you can refer to the data sheet.

12.ICSP pin

Mostly, ICSP (12) is an AVR, a tiny programming header for the Arduino consisting of MOSI, MISO, SCK, RESET, VCC, and GND. It is often referred to as an SPI (Serial Peripheral Interface), which could be considered as an "expansion" of the output. Actually, you are slaving the output device to the master of the SPI bus.

13.Power LED indicator

This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.

14.TX and RX LEDs

On your board, you will find two labels: TX (transmit) and RX (receive). They appear in two places on the Arduino UNO board. First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication. Second, the TX and RX led (13). The TX led flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.

15.Digital I/O

The Arduino UNO board has 14 digital I/O pins (15) (of which 6 provide PWM (Pulse Width Modulation) output. These pins can be configured to work as input digital pins to read logic values (0 or 1) or as digital output pins to drive different modules like LEDs, relays, etc. The pins labeled ~ can be used to generate PWM.

16.AREF

In an easy- AREF stands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

3.3 ULTRASONIC SENSOR

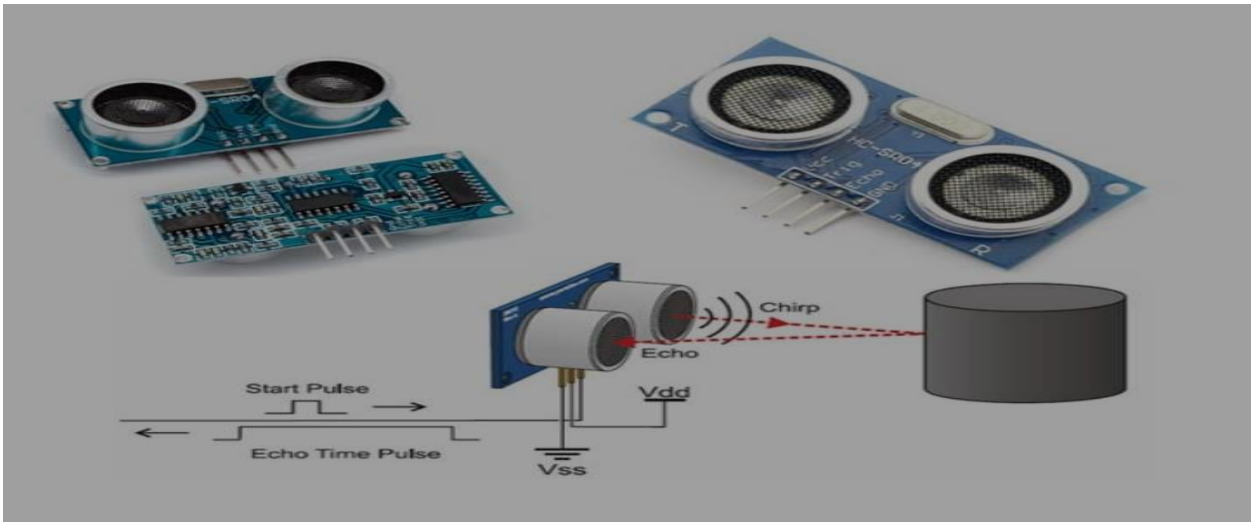

Fig.3.3.Ultrasonic Sensor

Ultrasonic is an excellent way of figuring out what's in the immediate vicinity of your Arduino. The basics of using ultrasound are like this: you shoot out a sound, wait to hear it echo back, and if you have your timing right, you'll know if anything is out there and how far away it is. This is called echolocation and it's how bats and dolphins find objects in the dark and underwater, though they use lower frequencies than you can use with your Arduino.

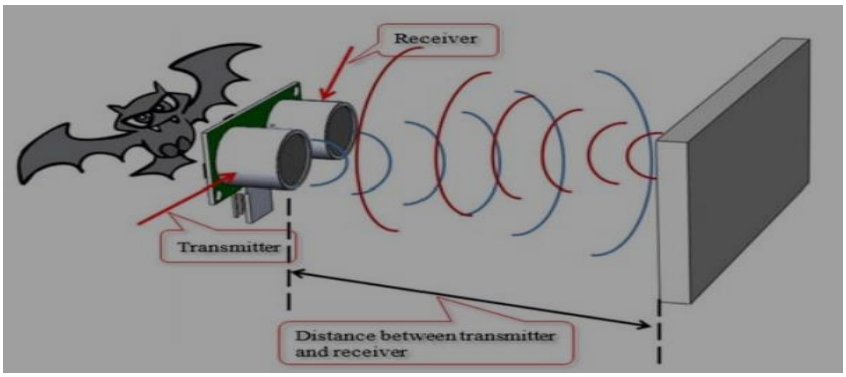

Fig.3.4.Distance Calculation

HC-SR04 Ultrasonic Sensor is a very affordable proximity/distance sensor that has been used mainly for object avoidance in various robotics projects. It has also been used in turret applications, water level sensing, and even as a parking sensor.

This module is the second generation of the popular HC-SR04 Low Cost Ultrasonic Sensor. Unlike the first generation HC-SR04 that can only operate between 4.8V~5V DC, this new version has wider input voltage range, allow it to work with controller operates on 3.3V.HC-SR04 ultrasonic sensor provides a very low-cost and easy method of distance measurement. It measures distance using sonar, an ultrasonic (well above human hearing) pulse (~40KHz) is transmitted from the unit and distance-to-target is determined by measuring the time required for the echo return. This sensor offers excellent range accuracy and stable readings to-use package. An on board 2.54mm pitch pin header allows the sensor to be plugged into a solderless breadboard for easy prototyping.

MODULATION SPECIFICATIONS

Electrical Parameters	Value
Operating Voltage	3.3Vdc ~ 5Vdc
Quiescent Current	<2mA
Operating Current	15mA
Operating Frequency	40KHz
Operating Range & Accuracy	2cm ~ 400cm (1in ~ 13ft) ± 3mm
Sensitivity	-65dB min
Sound Pressure	112dB
Effective Angle	15°
Connector	4-pins header with 2.54mm pitch
Dimension	45mm x 20mm x 15mm
Weight	9g

Table.3.1.Modulation specifications

HARDWARE INFORMATION

VCC	3.3v ~ 5V
TRIG	Triggering Input Pin. 10uS TTL Pulses
ECHO	TTL Logic Output Pin. Proportional to distance
GND	Ground Pin

Table.3.2.Hardware Information

LIQUID LEVEL DETECTION

Ultrasonic sensors are widely used for liquid level detection. In such cases, place a pipe on top of the sensor head as shown Figure-3. By detecting the liquid level inside the pipe, a wavy surface

or bubbles which can disturb stable reading can be prevented.

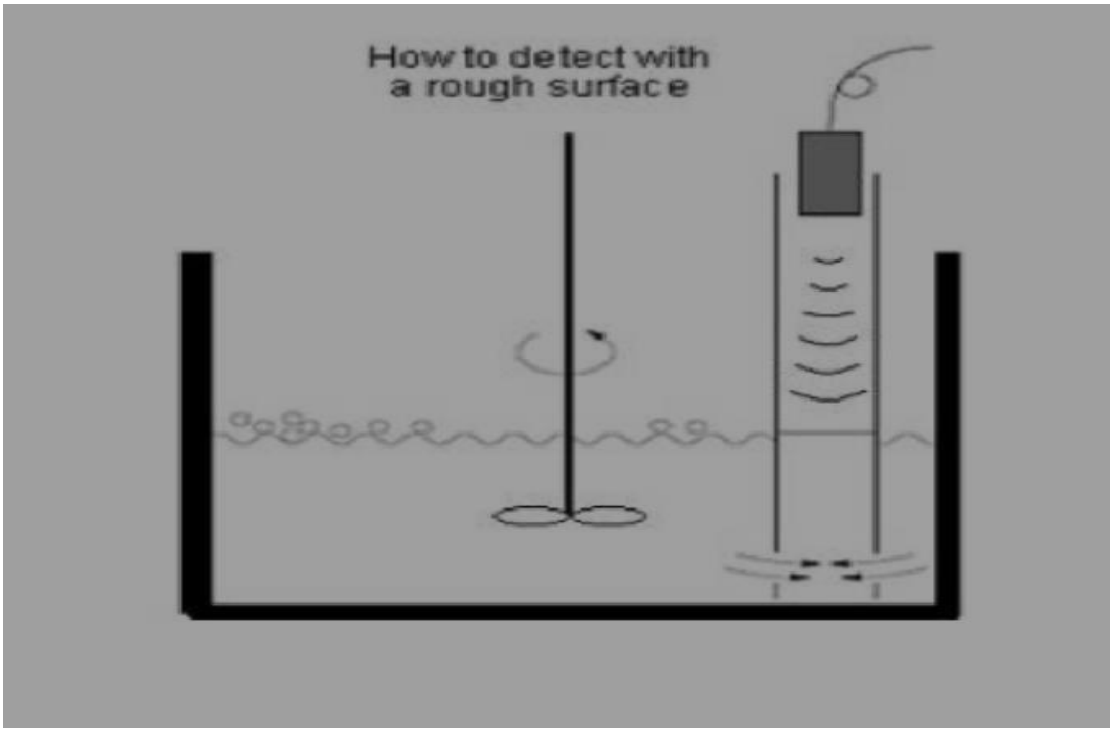

Fig.3.5.Detection of liquid level

3.4 LCD DISPLAY USING I2C

This is I2C interface 16x2 LCD display module, a high-quality 2 line 16 character LCD module with on-board contrast control adjustment, backlight and I2C communication interface. For Arduino beginners, no more cumbersome and complex LCD driver circuit connection. The real significance advantages of this I2C Serial LCD module will simplify the circuit connection, save some I/O pins on Arduino board, simplified firmware development with widely available Arduino library.

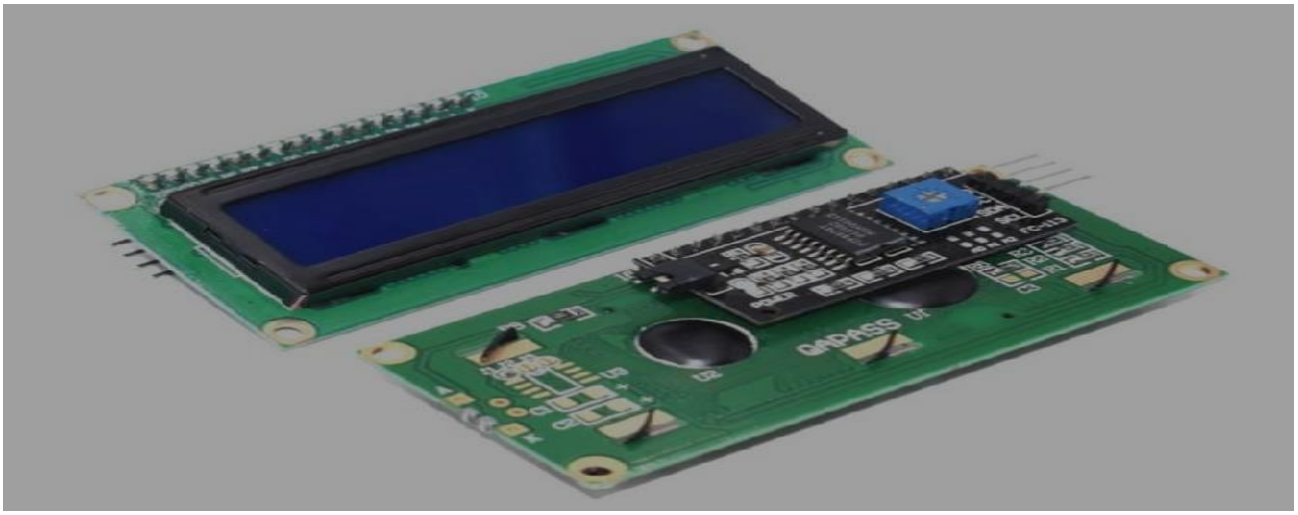

Fig.3.6.LCD Display

Brief Data:

• Compatible with Arduino Board or other controller board with I2C bus.

• Display Type: Negative white on Blue backlight.

• I2C Address:0x38-0x3F (0x3F default)

• Supply voltage: 5V • Interface: I2C to 4bits LCD data and control lines. • Contrast Adjustment: built-in Potentiometer.

• Backlight Control: Firmware or jumper wire. • Board Size: 80x36 mm.

I2C LCD Display.

At first you need to solder the I2C-to-LCD piggy-back board to the 16-pins LCD module. Ensure that the I2C-to-LCD piggy-back board pins are straight and fit in the LCD module, then solder in the first pin while keeping the I2C-toLCD piggy-back board in the same plane with the LCD module. Once you have finished the soldering work, get four jumper wires and connect the LCD

module to your Arduino as per the instruction given below.

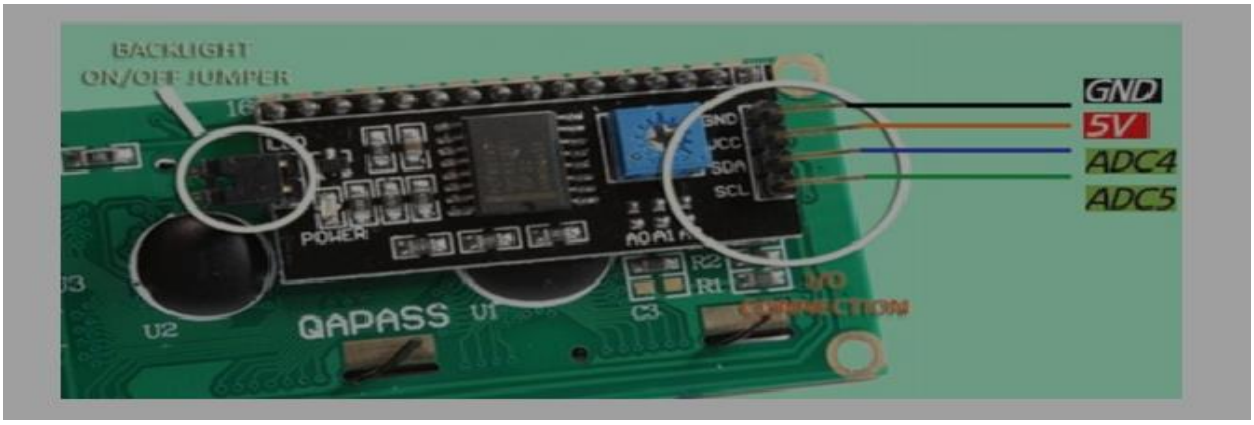

Fig.3.7.I2C Module

3.5 RELAY MODULE

1 Channel 5V Optical Isolated Relay Module

This is a LOW Level 5V 1-channel relay interface board, needs a 15-20mA driver current. It can be used to control various appliances and equipment with large current. It is equipped with high-current relays that work under AC250V 10A or DC30V 10A. It has a standard interface that can be controlled directly by microcontroller. This module is optically isolated from high voltage side for safety requirement and also prevent ground loop when interface to microcontroller.

Fig.3.8.Relay module

- Operating Voltage: 5Vdc.
- Relay Maximum output: DC 30V/10A, AC 250V/10A.
- 1 Channel Relay Module with Opto-coupler. LOW Level Trigger expansion board, which is compatible with Arduino control board.
- Standard interface that can be controlled directly by microcontroller (8051, AVR, *PIC, DSP, ARM, ARM, MSP430, TTL logic).
- Relay of high quality low noise relays SPDT. A common terminal, a normally open, one normally closed terminal.
- Opto-Coupler isolation, for high voltage safety and prevent ground loop with microcontroller.

Operating Principle:

See the picture below: A is an electromagnet, B armature, C spring, D moving contact, and E fixed contacts. There are two fixed contacts, a normally closed one and a normally open one. When the coil is not energized, the normally open contact is the one that is off, while the normally closed one is the other that is on. Supply voltage to the coil and some currents will pass through the coil thus generating the electromagnetic effect. So the armature overcomes the tension of the spring and is attracted to the core, thus closing the moving contact of the armature and the normally open (NO) contact or you may say releasing the former and the normally closed (NC) contact. After the coil is de-energized, the electromagnetic force disappears and the armature moves back to the original position, releasing the moving contact and normally closed contact. The closing and releasing of the contacts results in power on and off of the circuit.

Input:

VCC : Connected to positive supply voltage (supply power according to relay voltage)

GND : Connected to supply ground.

IN1: Signal triggering terminal 1 of relay module

Output:

Each module of the relay has one NC (normally close), one NO (normally open) and one COM (Common) terminal. So there are 2 NC, 2 NO and 2 COM of the channel relay in total. NC stands

for the normal close port contact and the state without power. NO stands for the normal open port contact and the state with power. COM means the common port. You can choose NC port or NO port according to whether power or not.

3.6 PUMP

This 12V 385 Water Pump can be used to water your plants, make a fountain or waterfall, and even change your fish tank water. It works quietly with the sound level under 30db. The pump has a filter inside as well as a rubber fixing soft bracket which can help secure it to surfaces.

Fig.3.9.water pump

- Pump Size: Ø35x80mm.
- Working voltage: 12VDC.
- Working current: 0.5~0.7A.
- Empty Load Current: 0.18A.
- Nozzle Diameter: Ø7mm.
- Flow Rate: 1.5~2L / Min (approx).
- Vertical Lift: Vertical up to 2 meters.
- Suitable Tubing Diameter: Ø6mm.

CHAPTER 4

PROJECT DESCRIPTION

4.1 OBJECTIVES

There are some objectives need to be achieved in order to accomplish this project. These objectives will act as a guide and will restrict the system to be implemented for certain situations:

1. To develop water level control system to control the water level in the tank.

2. To check the level of water in the tank. Depending on the water level, the motor switches ON when the water level goes below a predetermined level or the motor switches OFF when the tank is full.

3. To display the water level and other important data on a LCD Display.

4. To monitor the level of water in the tank. If the level inside the tank is low, the motor

4.2 CIRCUIT DIAGRAM

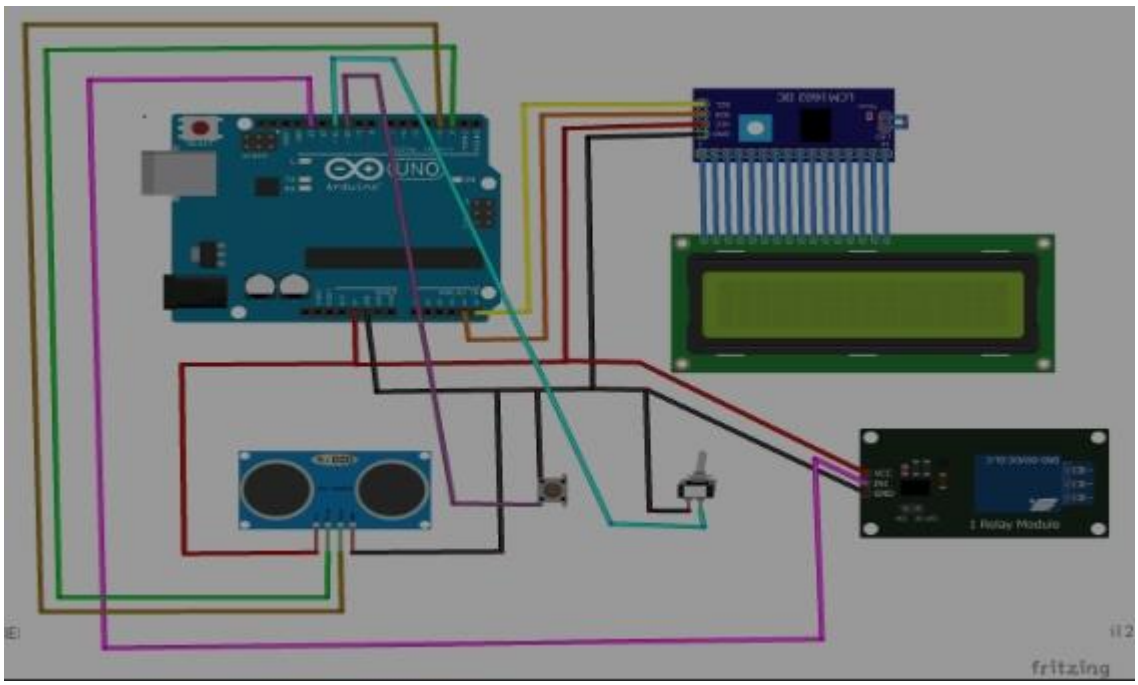

Fig.4.1.Circuit Diagram

4.3 CODE

```
#include <Wire.h>

#include <EEPROM.h>

#include <LiquidCrystal_I2C.h>

// Initialize the I2C LCD (Address: 0x27, 16 columns, 2 rows)

LiquidCrystal_I2C lcd(0x27, 16, 2);

long duration, inches;

int set_val, percentage;

bool state, pump;

void setup() {

  lcd.init();   // Initialize the LCD

  lcd.backlight();  // Turn on the backlight

  lcd.setCursor(0, 0);

  lcd.print("WATER LEVEL:");

  lcd.setCursor(0, 1);

  lcd.print("PUMP:OFF MANUAL");

  pinMode(8, OUTPUT);

  pinMode(9, INPUT);

  pinMode(10, INPUT_PULLUP);

  pinMode(11, INPUT_PULLUP);

  pinMode(12, OUTPUT);

  set_val = EEPROM.read(0);

  if (set_val > 150) set_val = 150;

}
```

```
void loop() {
   digitalWrite(8, LOW);
   delayMicroseconds(2);
   digitalWrite(8, HIGH);
   delayMicroseconds(10);
   digitalWrite(8, LOW);
   duration = pulseIn(9, HIGH);
   inches = microsecondsToInches(duration);
    percentage = (set_val - inches) * 100 / set_val;
   lcd.setCursor(12, 0);
   if (percentage < 0) percentage = 0;
   lcd.print(percentage);
   lcd.print("%   ");
   if (percentage < 30 && digitalRead(11)) pump = 1;
   if (percentage > 99) pump = 0;
   digitalWrite(12, !pump);
   lcd.setCursor(5, 1);
   if (pump == 1) lcd.print("ON ");
   else lcd.print("OFF");
   lcd.setCursor(9, 1);
 if (!digitalRead(11)) lcd.print("MANUAL");
   else lcd.print("AUTO   ");
 if (!digitalRead(10) && !state && digitalRead(11)) {
 state = 1;
```

```
 set_val = inches;
      EEPROM.update(0, set_val);
   }
     if (!digitalRead(10) && !state && !digitalRead(11)) {
      state = 1;
      pump = !pump;
   }
     if (digitalRead(10)) state = 0;
   delay(500);
}
long microsecondsToInches(long microseconds) {
   return microseconds / 74 / 2;
}
```

4.4 FLOW CHART

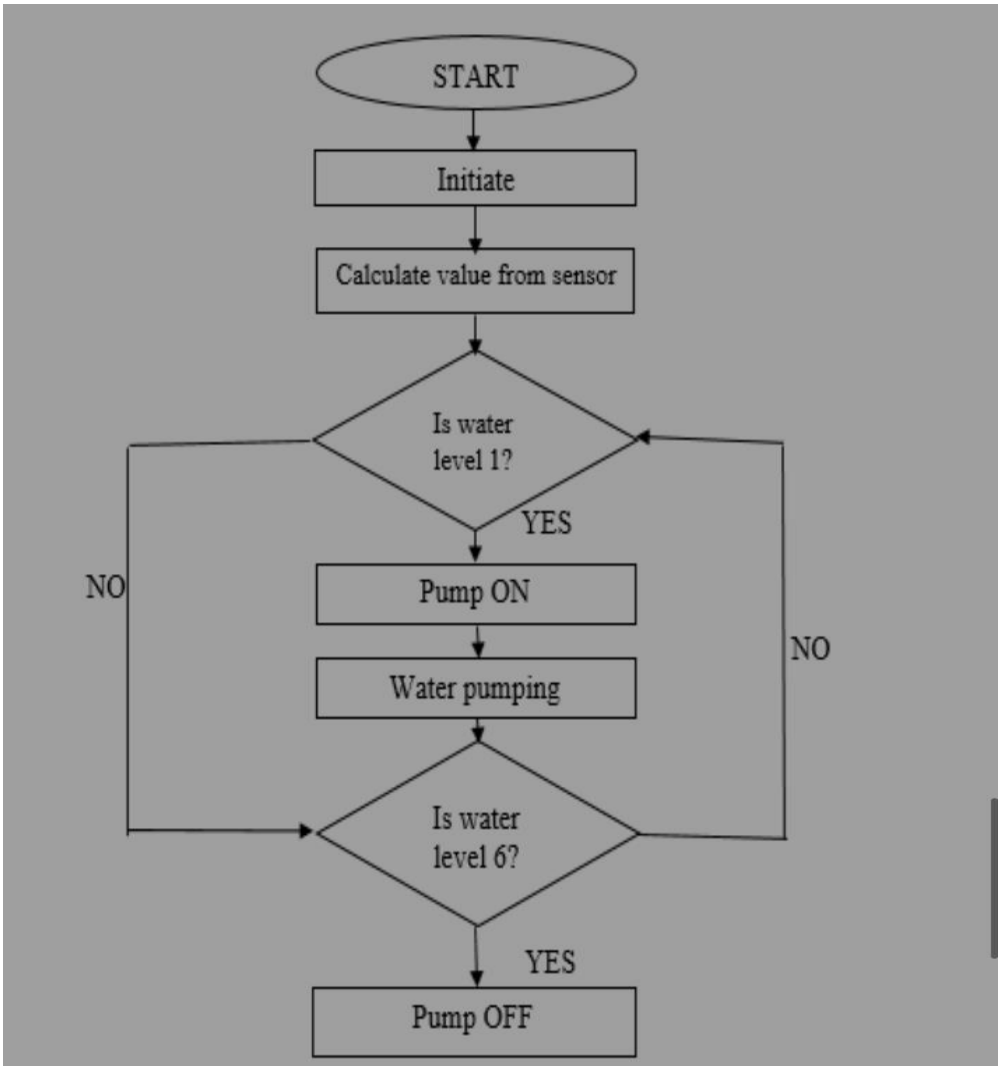

Fig.4.2.Flow chart

4.5 APPLICATIONS

Household Applications

1. **Overhead Water Tank Monitoring** – Automatically control motor to fill the tank and prevent overflow.
2. **Underground Sump Pump Control** – Manage water transfer from sump to overhead tank.
3. **Water Heater Level Monitoring** – Prevent dry heating by ensuring minimum water levels.

4. **Smart Garden Irrigation** – Ensure water tank for irrigation is always filled to avoid dry soil.
5. **Aquarium Water Management** – Maintain optimal water level in home aquariums.

Agricultural Applications

6. **Farm Irrigation System** – Automate irrigation pump based on tank level.
7. **Drip Irrigation Water Tank Control** – Ensure steady supply for drip irrigation systems.
8. **Water Storage for Livestock** – Monitor and refill water troughs or tanks for animals.
9. **Rainwater Harvesting System** – Control tank levels to avoid overflow or dry conditions.
10. **Greenhouse Water Management** – Keep water levels stable for plant beds or misting systems.

Industrial Applications

11. **Chemical Tank Monitoring** – Monitor and control liquid levels in chemical storage tanks.
12. **Cooling Tower Water Level Control** – Maintain levels for industrial cooling systems.
13. **Boiler Feedwater Management** – Prevent boiler damage due to low water levels.
14. **Beverage Production** – Monitor ingredient water levels in manufacturing.
15. **Water Treatment Plants** – Automate tank filling and drainage processes.

Commercial & Institutional Applications

16. **Hotel or Apartment Water Supply** – Maintain water levels in shared building tanks.
17. **School/College Campuses** – Monitor and manage water usage efficiently.
18. **Public Restrooms & Fountains** – Automate refill mechanisms for water tanks.
19. **Swimming Pool Water Control** – Maintain proper pool water levels.
20. **Emergency Water Supply Systems** – Ensure backup tanks are always full and ready.

CHAPTER 5

RESULT AND DISCUSSIONS

Result:

The Arduino-based water level indicator and controller system was successfully designed and implemented. The system accurately detected different water levels in the tank using sensor probes and displayed real-time status on an LCD. It automatically activated the water pump when the water level dropped below the minimum threshold and turned it off once the tank was full. The system operated efficiently with minimal human intervention, demonstrating reliable automation and water conservation. It also included safety features such as a buzzer alert and dry-run protection, validating its effectiveness for both residential and industrial applications.

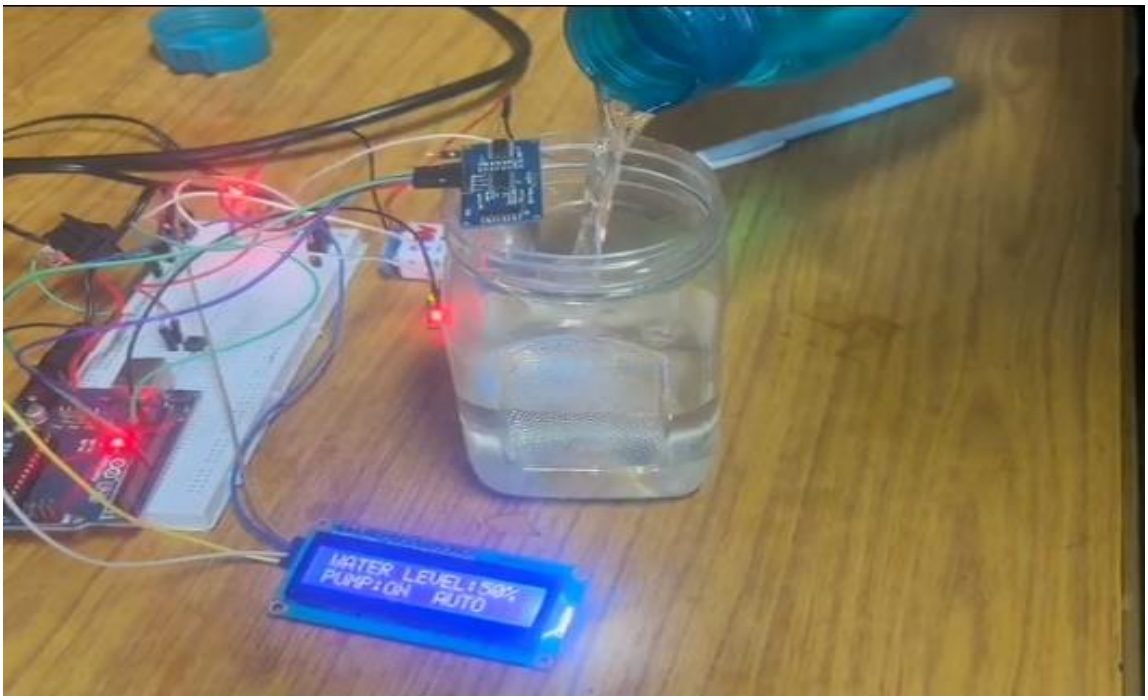

Fig.5.1.water level indicator and controller

Discussion:

The development of a water level indicator and controller using Arduino highlights the growing importance of automation in resource management. During the project, several aspects were considered, including sensor accuracy, pump control logic, user interface (LCD display), and system reliability. The use of Arduino allowed for easy integration of sensors and components, as well as simple programming for control logic.

One of the key challenges faced was ensuring accurate water level detection, especially when using conductive-type sensors, which may be affected by water impurities or corrosion over time. Calibration and proper spacing of sensor probes helped to mitigate this. Another consideration was the safety of the pump operation. To avoid dry-run conditions, the system was programmed to prevent the pump from turning on if the lower tank was empty, showcasing the importance of adding protective logic in embedded systems.

The project also opens up discussions about scalability. While the current design is ideal for small to medium-scale applications, industrial-level systems may require more robust sensors and wireless communication for remote monitoring. Furthermore, adding IoT capabilities (like Wi-Fi or GSM modules) could enable users to monitor and control the system remotely through smartphones or the internet.

Overall, the project provides a strong foundation for smart water management and can be extended with advanced features based on real-world needs and applications.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Conclusion:

This project effectively demonstrates the practical application of Arduino in automating water level management. By integrating basic electronic components and sensors, the system provides a cost-effective, efficient, and user-friendly solution to common water usage challenges. It not only reduces manual monitoring and labor but also helps in conserving water and protecting electrical appliances like water pumps. The design is scalable and can be enhanced with IoT features for remote monitoring. Overall, the project contributes to smarter resource management and supports sustainable water usage practices.

Future Scope:

The current implementation of the Arduino-based water level indicator and controller serves as a reliable and low-cost solution for water management. However, there is significant potential to enhance its functionality and expand its applications in the future. One of the most promising areas of improvement is the integration of **IoT (Internet of Things)** features, enabling users to monitor and control water levels remotely through a smartphone app or web interface. This would be especially useful for large residential buildings, farms, and industries.

Additionally, **wireless sensors** can be used to eliminate the need for physical probe connections, reducing maintenance and improving system durability. The inclusion of **data logging** features to track water usage over time can also provide valuable insights for conservation efforts. For larger systems, incorporating **solar power** can make the project more sustainable and energy-efficient. Moreover, adding **voice assistant compatibility** (e.g., with Alexa or Google Assistant) could provide a hands-free way to monitor tank status or control the motor. These advancements would make the system more intelligent, adaptive, and scalable for diverse applications in smart homes, smart agriculture, and urban infrastructure.

Bibliography

[1] Asaad Ahmed Mohammed ahmed Eltaieb and Zhang Jian Min, Automatic Water Level Control System, International Journal of Science and Research (IJSR), Volume 4 Issue 12, December 2015.

[2] Beza Negash Getu and Hussain A. Attia, Automatic Water Level Sensor and Controller System, Â©2016 IEEE.

[3] Priya J, Sailusha Chekuri, water level monitoring system using IoT, International Research Journal of Engineering and Technology (IRJET) Volume: 04 Issue: 12, Dec-2017.

[4] Madhurima Santra, Sanjoy Biswas, Sibasis Bandhapadhyay and Kaushik Palit, Smart Wireless water level Monitoring & Pump controlling System, International Journal of Advances in Scientific Research and Engineering (IJASRE), Vol. 03, Issue 4, May -2017.

[5] K. Santhosh Kumar, G. Mukesh, K. Deepti, Microcontroller based Automatic Water level Control System ,International Refereed Journal of Engineering and Science (IRJES) Volume 4, Issue 11, November 2015.

[6] Sanam Pudasaini, Anuj Pathak, Sukirti Dhakal and Milan Paudel, “Automatic Water Level Controller with Short Messaging Service (SMS) Notification”, International Journal of Scientific and Research Publications, Volume 4, Issue 9, September 2014.

Appendix

This project utilizes an Arduino Uno microcontroller to build an automatic water level indicator and controller system. The key components used in this project include an ultrasonic sensor (HC-SR04), a 1-channel relay module, a submersible water pump, three LEDs (red, yellow, and green), resistors, jumper wires, a breadboard or PCB, and a power supply. The ultrasonic sensor is positioned at the top of the water tank to measure the distance to the water surface. This distance is continuously monitored by the Arduino, which processes the data to determine the water level within the tank.

The system operates based on three defined levels—low, medium, and high. When the water level falls below a predefined threshold (e.g., greater than 25 cm from the sensor), the red LED turns on and the relay activates the pump to fill the tank. When the level reaches a medium range (between 10 cm and 25 cm), the yellow LED lights up, indicating that the tank is partially filled, and the pump is turned off. If the water level rises to within 10 cm of the sensor, the green LED turns on, signaling that the tank is full, and the pump remains off to prevent overflow.

The control logic is written in Arduino C++ and includes a simple trigger and echo mechanism to measure the distance using the ultrasonic sensor. The pump is controlled through a relay module connected to one of the Arduino's digital output pins. LED indicators are connected to separate digital pins and are used to visually represent the water level in real time.

During testing, the system was calibrated using actual measurements of the water tank depth, and the threshold distances were adjusted accordingly to ensure accurate readings. The sensor was found to be reliable with a response delay of approximately one second. The relay module responded correctly to the control signals from the Arduino, and the LEDs clearly indicated the current water level.

This water level monitoring system is highly useful for home water tanks, especially in areas where water supply is inconsistent or requires manual monitoring. It helps in conserving water, avoiding overflow, and protecting pumps from running dry. The project can be further enhanced by integrating an LCD display or by adding IoT functionality for remote monitoring and control.